One tuning for the whole operating envelope: split-range PID for the zones of a rubber-line temperature control unit

Systems engineering dossier. Pre-baseline draft, revision A, 11 Nov 2026.

This dossier follows C. S. Wasson, *System Engineering Analysis, Design, and Development* (2nd ed., Wiley 2016), cited as **[W]** with the printed page and then the PDF page. Reading notes are in `docs/sources/`. Other sources are listed in section 15 and in `docs/sources/REFERENCES.md`, each with its status: **[read]**, **[std]** (textbook, section named), **[derived]** (computed here) or **[assumed]** (a sizing choice of this repo, listed in section 8.2 with its sensitivity).

The engineering work is in this repository:

- the zone physics (`src/tcutune/physics.py`, `plant.py`), identification and tuning tool (`identify.py`, `tuning.py`, `tool.py`, CLI `tcu-tune`);
- the cooling linearisation and the envelope analysis (`linearise.py`, `envelope.py`);
- the PLC code in `plc/` (SCL, compiled in TIA Portal V18);
- the study `scripts/run_study.py` → `results/study.md`, and the trade study `scripts/aoa.py` → `results/aoa.md`.

> **Status.** Everything is simulation plus a TIA Portal V18 compile. No data from a real TCU or line has been used, and no PLCSIM run has been made. The open items that need the plant are in section 12.

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0. Summary of findings

All numbers are from simulation (section 8): four zones tuned by the tool from simulated technician tests at one test point (80 °C, full process load, cooling water 15 °C), then checked on the true linearised plant at about 150 envelope points per zone (setpoint 50–90 °C, load 0.5–1.5 × q_run, cooling water 8–30 °C), clean and fouled, and in the time domain at a winter and a summer corner. Source: `results/study.md` (sections in brackets) and `results/aoa.md`.

1. **The engineering question.** A split-range PI per zone holds well at some conditions and badly at others. What moves under the controller, and what is the smallest change that keeps one tuning robust (SPS-201: Ms ≤ 1.6) over the whole envelope?

2. **What moves (§4, the deciding parameter).** The cooling-path gain near the loop's crossover frequency, relative to the test point, spans **0.07–1.87**: a factor of **13.7–25** on zones with a linear (PWM) valve and **3.5–3.6** with an equal-percentage valve. Causes: the heat removed scales with Tw − Tcw (13–82 K over the envelope), and the plate exchanger's ε-NTU curve bends the valve characteristic. The heating path does not move.

3. **Alternatives (§5, §3).**

Alternative	Ms in cooling, clean (all zones)	points with Ms > 2	summer load-step peak, zones A/B/D
A one gain set	1.08–298	53–86 %	0.9 / 1.5 / 2.2 K, but limit cycle (707 switchovers, zone A test point)
B static gain ratio	1.07–149	41–99 %	—
C crossover ratio (v0.1)	1.02–1.70	0 % (9–39 % below 1.2)	2.9 / 2.9 / 4.2 K
D C + Tw − Tcw scaling	1.09–3.35	0–9 %	—
E C + exchanger/valve inversion (v0.2)	**1.36–1.41**	0 %	**1.7 / 1.1 / 2.2 K**

   With the exchanger fouled and the model left clean, E stays at **1.19–1.33**: fouling only makes it more cautious.

4. **Why the obvious fix fails.** On the equal-percentage zones the static gain ratio is below 1 (0.49, 0.75): the open valve adds conductance and lowers the static gain, while near crossover the cooling path is 2.4–2.6 × stronger than heating. B is worse than doing nothing there.

5. **A defect found by the corner runs.** With E, the valve can saturate before the PID's own cooling limit; the integrator wound up and the summer setpoint drop overshot by 1.6–1.8 K. Conditional integration from `FB_CoolLinearise.Saturated` brought it to 0.2–0.4 K (SPS-205).

6. **Feedforward (§6).** The mixer's 150 s batch cycle sits where feedback amplifies (|S| = 1.25; |S| > 1 for 38–224 s), so no retuning can attenuate it (Bode sensitivity integral [C3]). Feedforward from the rotor power halves the ripple, **0.8 → 0.4 K** peak-to-peak (0.4–0.5 K with ±30 % gain error). On the head zone with the sensor in the steel no cooling alternative meets the 2 K load-step limit (6.5 K with E); feedforward brings it to **2.6 K** (2.3–3.4 K with ±30 % error).

7. **Trade study (§10).** E 80.2, C 77.4, D 43.9, B 35.5, A 33.7. The margin is small: C wins when cost weighs 27 or more, or when the sensor and engineering cost doubles. Recommendation: **retune all zones with C now (no hardware); add the cooling-water sensor and E where the cooling water swings with the season or the valves are equal-percentage; feedforward on mixer and head zones.**

8. **PLC (§7).** All blocks compile in TIA Portal V18 with 0 errors and 0 warnings, OB30 at 100 ms; upper-bound 0.64 ms per zone per cycle, 15.4 ms for 8 zones with a 3× margin (15 % of the cycle).

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1. Problem and stakeholders ([W] ch. 4–5)

1.1 Problem statement

A rubber-compound line (internal mixers, cold-feed extruders, their heads and the conditioning stations) holds the temperature of its machine parts with **temperature control units (TCUs)**. A TCU has several independent water circuits (zones). Each zone heats with an electric heater and cools through a cooling-water valve and a plate heat exchanger in its closed primary circuit. One PID per zone, split-range between heater and valve, runs in a Siemens S7 PLC.

Problem statement in the form of [W] p.91–92/121–122 (one sentence, no cause, no solution, no blame):

> *Some TCU zones on the line hold their temperature well at some operating conditions and oscillate or respond slowly at others, and there is no method for setting their controllers that works for all zones and all conditions.*

The plant's own description says the current temperature control "is showing non-optimal behaviour in some cases". The phrase "in some cases" is the starting point: a tuning that is bad everywhere is easy to find and fix; one that is bad *somewhere* means the plant moves under the controller.

1.2 Stakeholders ([W] Table 4.2, p.84–85/114–115)

Role ([W] Table 4.2)	Who	Interest
User	line operators, process engineers	stable compound temperature, few alarms, quick start-up
End user	tyre building (the compound's customer)	consistent compound properties (temperature history affects scorch safety and viscosity)
Owner	plant	scrap, energy, availability
Maintainer	automation and maintenance technicians	one method for all TCUs, easy retuning after a repair, diagnosable faults
Developer	thesis student, automation group	PLC code that fits the existing S7 programs and cycle
Independent test	process engineering, quality	acceptance on the real line against the current settings

1.3 Needs, measures of effectiveness and measures of performance ([W] p.100–101/130–131, Fig. 5.8 p.116/146)

Need	MOE ([W] P5.10)	Contributing MOPs
N1 compound temperature held in production	share of production time inside ±1 K	peak deviation after a load change; IAE;
N2 the same tuning works in every condition	worst-case robustness over the operating envelope	Ms in heating and in cooling at every envelope point, exchanger clean and fouled
N3 no oscillation, no valve or contactor wear	switchovers heater ↔ valve per hour	limit-cycle detection, valve travel
N4 one tool for all similar TCUs	technician time per zone	test duration, number of tests, no manual model fitting
N5 fits the existing PLC	no new hardware except where the analysis shows it pays	PLC time per zone; new signals

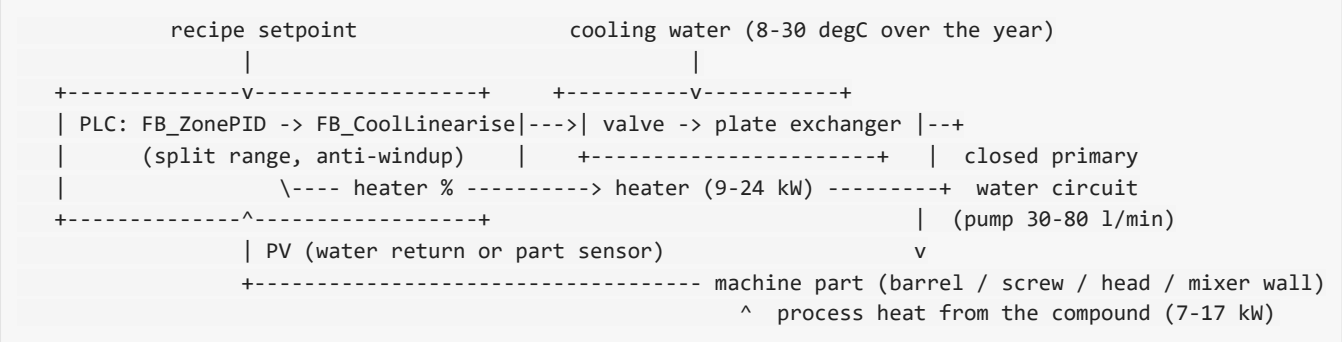
MOS ([W] P5.11, p.115/145): the zone stays in automatic through the whole operating envelope without operator retuning.

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2. Concept of operations ([W] ch. 6)

2.1 Mission and context

The mission of one TCU zone controller is to hold the temperature of its machine part at the setpoint given by the recipe, across product changes, start-ups, process load changes and the seasons, with the heater and the cooling valve it has, inside the PLC cycle it has.



The technician's part of the mission is the **tuning tool** (`tcu-tune`): three short tests per zone (heater step, cooling step with the machine running, optional relay test), recorded as CSV from a trace, give the recommended `FB_ZonePID` and `FB_CoolLinearise` settings.

2.2 Phases and use cases ([W] P5.7 p.110/140; UC principles p.120–122/150–152; Table 6.4 p.143/173)

Phase	Use case	Actor	Trigger	Outcome
Pre-mission	UC1 commission or retune a zone	technician	new TCU, repair, recipe range change	tests recorded, settings loaded, acceptance run done
Pre-mission	UC2 enter the TCU nameplate data	technician	first commissioning of a TCU type	exchanger UA, flows, valve type in `FB_CoolLinearise`
Mission	UC3 cold start to setpoint	operator	shift start	setpoint reached without overshoot beyond limit
Mission	UC4 production with process load	line	extruder/mixer running	zone in cooling, within ±1 K
Mission	UC5 recipe change (setpoint step)	recipe system	product change	new setpoint, bounded overshoot
Mission	UC6 batch load (mixer)	mixer cycle	every 2–4 min	periodic load absorbed

Phase	Use case	Actor	Trigger	Outcome
Mission	UC7 sensor or actuator fault	PLC	plausibility	defined degraded mode (section 3)
Post-mission	UC8 cleaning of the exchanger	maintenance	fouling, valve near 100 %	UA restored, no retuning needed

3. Modes and states ([W] ch. 7)

Modes follow [W] Table 7.3 (p.162/192) and the transition table form of Table 7.1 (p.158/188).

Mode	What runs	Entered by	Left by
M0 Off / Reset	outputs 0, integrator cleared	`Reset`	`Reset` released
M1 Manual	output = `ManualValue`, integrator tracks (bumpless)	`ManualEnable`	`ManualEnable` released
M2 Auto, heating	split range u ≥ 0 → heater	u ≥ deadband	u < −deadband
M3 Auto, cooling	split range u < 0 → cooling demand → linearisation → valve	u ≤ −deadband	u > deadband
M3a Cooling saturated	valve 100 %, PID stops integrating into cooling (conditional integration)	`FB_CoolLinearise.Saturated`	demand back inside capacity
M3b Cooling-water sensor fault	last valid Tcw held	`TcwValid` = FALSE	sensor valid
T Test (commissioning)	tool test sequences (manual steps, relay)	technician	test end

Prohibited actions ([W] P7.10, p.169/199): no heater and valve on at the same time (the split range has one sign); no integration into a saturated valve; no change of `Mode`/ratio while in cooling without going through Manual.

4. Requirements ([W] ch. 19–22)

Requirements are written in the style of [W] ch. 20–22 (one per statement, verifiable, solution-free where possible; P19.6, p.401/431) with verification methods I/A/D/T ([W] p.270–272/300–302). Values marked **TBD** need the plant (section 12).

4.1 Mission and operational performance (spec §3.1–3.2)

ID	Requirement	Threshold / objective	Source	Verif.
SPS-101	In steady production the zone temperature shall stay within ±1 K of the setpoint.	≥ 99 % of time / ≥ 99.9 %	N1; TCU control accuracy class ±0.5–1 K [assumed], TBD-1 with the plant	A, T
SPS-102	After a process-load step of q_run the peak deviation shall not exceed	2.0 K / 1.5 K	N1 [assumed]	A, T
SPS-103	A setpoint step of −5 K in the cooling range shall overshoot by no more than	1.0 K / 0.5 K	N1, UC5 [assumed]	A, T
SPS-104	In steady production there shall be no sustained heater ↔ valve limit cycle (valve cycling between closed and fully open).	none / none	N3, contactor and valve life	A, T
SPS-105	With a batch load (mixer), the periodic temperature ripple shall not exceed	±0.5 K / ±0.25 K	N1, UC6 [assumed]	A, T

4.2 Capabilities (spec §3.3)

ID	Requirement	Threshold / objective	Source	Verif.
SPS-201	One set of controller settings per zone shall keep Ms ≤ 1.6 (objective 1.4 ± 0.1) in heating and in cooling at every point of the operating envelope: setpoint 50–90 °C, process load 0.5–1.5 × q_run, cooling water 8–30 °C, exchanger clean to fouled (Rf = 0.000176 m^2K/W).	Ms ≤ 1.6 / 1.3–1.5	N2 [C1: Ms 1.2–2.0 usual]	A

ID	Requirement	Threshold / objective	Source	Verif.
SPS-202	The tuning tool shall produce the settings from a heater step, a cooling step with the machine running and an optional relay test, without manual model fitting.	≤ 3 h test time per zone	N4	D
SPS-203	The tool shall refuse rules whose model assumptions do not hold (e.g. FOPDT rules when no dead time is visible).	always	N4	T
SPS-204	Manual ↔ automatic shall be bumpless.	step ≤ 0.5 %	N3 [C1]	T
SPS-205	When the cooling valve is fully open before the PID reaches its own cooling limit, the PID shall stop integrating into cooling.	always	SPS-103 (windup)	T
SPS-206	A cooling-water temperature sensor fault shall not move the valve.	hold last valid value	FMEA F3	T

4.3 Interfaces (spec §3.4)

ID	Requirement	Source	Verif.
SPS-301	The controller shall use the existing zone temperature, the TCU water temperature, heater and valve outputs, and at most **one new signal per TCU**: the cooling-water supply temperature.	N5	I
SPS-302	Optional feedforward shall use a process-power signal that the line PLC already has (extruder drive power, mixer rotor power).	N5	I

4.4 Operating environment (spec §3.5)

Cooling water from a cooling-tower circuit, 8–30 °C over the year [assumed, TBD-2]; ambient 15–35 °C; setpoints 50–90 °C (water TCU without pressurisation above 95 °C [assumed]).

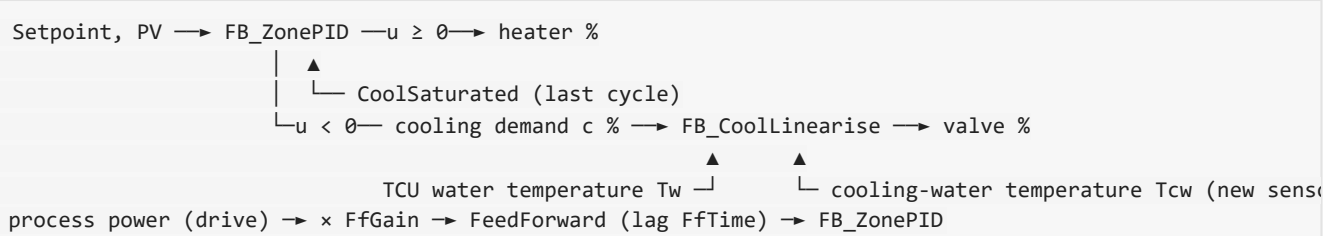
4.5 Design and construction constraints (spec §3.6)

ID	Requirement	Verif.
SPS-361	Siemens S7-1500, SCL, TIA Portal V18: 0 errors and 0 warnings.	T (compile)
SPS-362	Execution of all zone blocks shall stay below 20 % of the 100 ms OB30 period with a 3× margin ([W] p.669).	A, T (CPU statistics)
SPS-363	The SCL source executed in CI shall match the Python reference (bit-exact for FB_ZonePID; ≤ 0.01 % valve for FB_CoolLinearise; ≤ 0.05 K closed loop).	T

5. Architecture ([W] ch. 26)

The architecture separates what changes with the plant from what changes with the zone ([W] ch. 26, functional allocation). One zone is three PLC blocks plus the simulation blocks used for tests:

Element	Function	Allocated requirements
`FB_ZonePID` (v0.2)	split-range PI(D) in heating-equivalent %: setpoint weighting, filtered derivative on the measurement, back-calculation anti-windup, bumpless manual, **lagged feedforward input**, **conditional integration when the cooling valve saturates**, `CoolGainRatio`	SPS-101…105, 204, 205
`FB_CoolLinearise` (new)	turns the PID's cooling demand into a valve opening: mode 0 none, mode 1 Tw−Tcw scaling, **mode 2 inversion of the valve + plate-exchanger model (ε-NTU) by bisection**; holds the last valid Tcw on a sensor fault; reports saturation	SPS-201, 205, 206, 301
`tcu-tune` (Python tool)	identification (FOPDT, lead-lag with dead time, relay), rule selection with Ms check, crossover cooling ratio; with mode 2 active the cooling test is recorded through the linearisation	SPS-202, 203
`FB_ZonePlant`, `FB_Scenario`, `FB_ZoneTest`, OB30	simulation of a zone in the PLC (no hardware) for PLCSIM and CI equivalence tests	SPS-363



Why the linearisation sits outside the PID: the PID stays the plant's standard split-range block (retuning keeps its meaning), and the valve/exchanger knowledge, which belongs to the TCU type, is entered once per TCU type from the nameplate (UC2).

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6. Interfaces ([W] ch. 27)

6.1 Ownership

The zone blocks are owned by the line PLC program. The tool is owned by maintenance. The TCU nameplate data are owned by the TCU supplier documentation.

6.2 Interface table

ID	From → to	Signal	Type, unit, range	Rate	Status
IF-1	zone sensor → FB_ZonePID.Input	zone temperature (water return or steel)	Real, °C, 0–150, 0.1 K resolution	100 ms	existing
IF-2	TCU water sensor → FB_CoolLinearise.Tw	primary-circuit water temperature	Real, °C	100 ms	existing in every TCU (its own control sensor)
IF-3	new Pt100 → FB_CoolLinearise.Tcw (+ TcwValid)	cooling-water supply temperature	Real, °C, 0–50; validity from the AI channel diagnostics	100 ms (changes in hours)	**new, one per TCU**
IF-4	FB_ZonePID.OutHeat → heater	heater demand	Real, %, 0–100 → SSR/thyristor or PWM contactor	100 ms	existing
IF-5	FB_CoolLinearise.Valve → valve	valve opening	Real, %, 0–100 → proportional valve or PWM solenoid	100 ms	existing
IF-6	drive → FB_ZonePID.FeedForward	process power × FfGain	Real, heating-equivalent %	100 ms	existing signal, new use (optional)
IF-7	FB_CoolLinearise.Saturated → FB_ZonePID.CoolSaturated	saturation flag	Bool	100 ms	internal
IF-8	trace (CSV) → tcu-tune	t_s, heat_pct, cool_pct, pv_degC	CSV, 1 s	per test	tool input
IF-9	tcu-tune → DB	Gain, Ti, Td, CoolGainRatio, Mode, DTref, UA, CsMax, Cp, ValveR, FfGain, FfTime	Real	per commissioning	tool output

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7. Performance budgets and margins ([W] ch. 31)

7.1 PLC cycle-time budget ([W] P31.4 apportionment, p.668; frame-based systems, p.679/709)

Static upper bound: every arithmetic and comparison operator in the SCL body, FOR bodies multiplied by their trip count, all branches counted, × the CPU 1513-1 PN typical times [P1] (256 ns per float operation, 48 ns per word operation), EXP and LN counted as 20 float operations each [assumed]. Source: `results/study.md` §7.

Block	float ops	word ops	time
FB_ZonePID	52	45	15.5 µs
FB_CoolLinearise (mode 2, 16 model evaluations)	2365	455	627 µs
per zone			**0.64 ms**
8 zones with a 3× margin ([W] p.669)			**15.4 ms = 15 % of the 100 ms OB30** (SPS-362: ≤ 20 %)

The linearisation dominates. If a CPU runs short, the inversion can run every 5th cycle (Tw and Tcw change over tens of seconds) or with 12 instead of 16 evaluations (valve resolution 0.05 %), cutting it by 80 % or 25 %. Neither is needed at the estimated load. On-target timing is VT-7.

7.2 Temperature error budget at a load step (zone B, test point, E)

Contribution	Value	Source
sensor resolution	0.1 K quantisation	assumed AI card
noise	0.05 K 1σ	assumed
dynamic deviation after a full load step	0.9 K peak	`results/study.md` §3
total against the 2 K limit (SPS-102)	≈ 1.0 K	margin about 1 K

7.3 Robustness margin ([W] P31.5)

Item	Allocation	Achieved with E (clean / fouled)	Source
Ms in heating	≤ 1.4 (tool target)	1.36–1.40	§3
Ms in cooling over the envelope	≤ 1.6 (SPS-201)	1.36–1.41 / 1.19–1.33	§5
model error of the linearisation	stay ≤ 1.6	exchanger UA 47 % below the model (fouled): Ms moves 1.36–1.41 → 1.19–1.33, i.e. towards caution	§5

8. Modelling and simulation ([W] ch. 33)

8.1 Intended use and fidelity ([W] P33.1, p.706/736; fidelity matrix, p.708/738)

Intended use: compare cooling-path alternatives and tuning rules on their robustness over the operating envelope, and produce realistic step-test records for the tool. **Not** intended for: absolute temperatures of a real line, valve sizing, or proving SPS-101 (that needs the line trial, section 11.3).

Effect	Modelled	How	Fidelity
water and steel heat capacity	yes	two lumped masses	medium: one steel mass per zone, no axial gradient
channel film coefficient	yes	Gnielinski at the design flow [H1 eq. 8.62] + steel conduction in series	medium
plate exchanger	yes	ε-NTU counterflow, cooling-side film ∝ flow^0.8 [H1, H2]	medium
cooling-water valve	yes	linear (PWM solenoid) or equal-percentage R = 30 [V1]	medium; no hysteresis, no PWM ripple
transport delay, sensor lag	yes	plug-flow volume / pump flow; first-order lag	medium
process heat from the compound	yes, as an input	step (extruder) or square-wave batch cycle (mixer)	low: amplitude assumed, section 8.2
fouling	yes	UA / (1 + U·Rf) [H3]	order of magnitude
sensor noise and resolution	yes	0.05 K white noise, 0.1 K quantisation	typical analog input card
property change with temperature	no	properties at 80 °C	a few % on UA over 50–90 °C

Effect	Modelled	How	Fidelity
boiling, pressurisation, pump curve	no	—	outside the 50–90 °C range

Verification of the model itself (tests): steady state against a 1 h simulation (< 1e-6 K), Euler step 0.1 s against 0.01 s (< 0.02 K), linearisation against the numerical DC gain, closed-form FRF against the matrix FRF (< 2e-3), and the SCL twin FB_ZonePlant within 0.01 K of the Python model.

8.2 Sourced vs assumed parameters

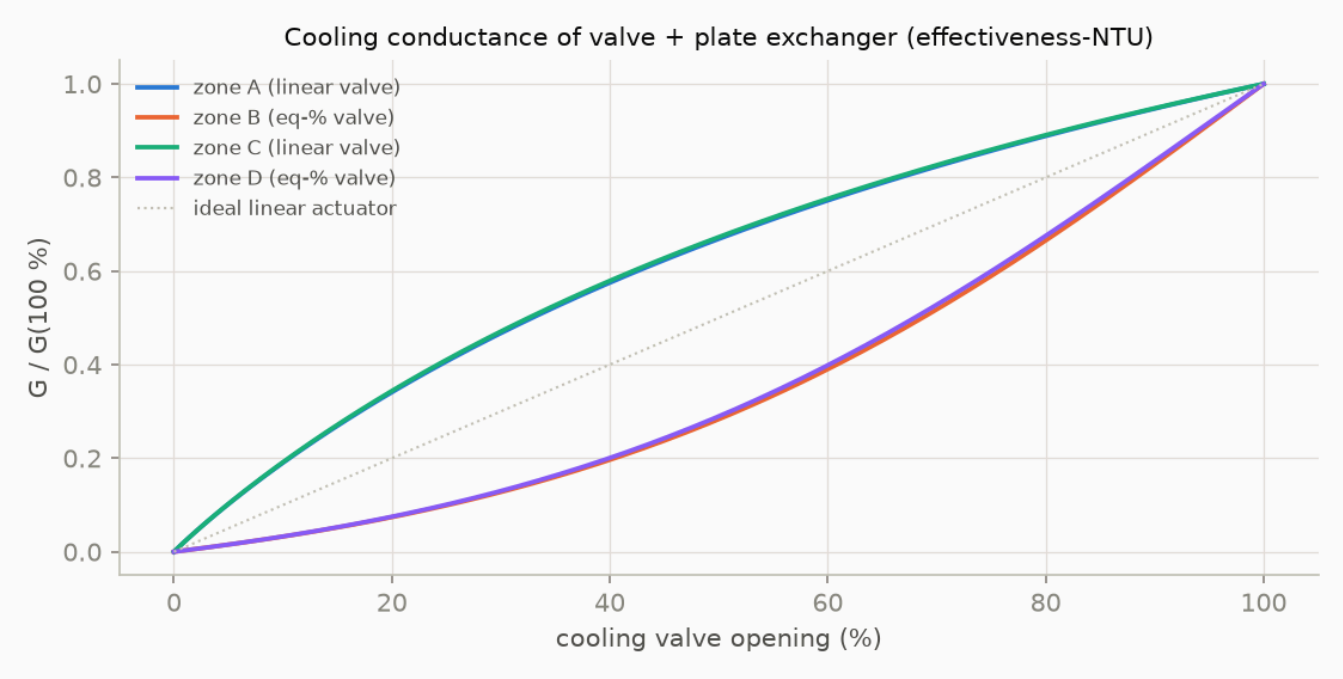
Parameter	Value(s)	Status	Sensitivity run
water properties, steel cp and k	Table A.1/A.6	[std] [H1]	—
Gnielinski, ε-NTU, Re^0.8	relations	[std] [H1, H2]	—
heater power	9 / 12 / 18 / 24 kW	[assumed] water-TCU class	the tool re-identifies, so only ratios matter
pump flow	30–80 l/min	[assumed]	enters L and Cp
exchanger UA at full cooling flow	0.6–2.6 kW/K	[assumed], sized so q_run needs 9–40 % valve	fouled case (UA −47 %)
valve type	A, C linear (PWM); B, D equal-% R = 30	[assumed]	both types are in the set
hoses, channels, steel mass	`physics.PRESET_DESIGNS`	[assumed] machine class	four zones span Cm 87–414 kJ/K
outer film 11 W/m²K		[std] estimate [H1 ch. 9]	small effect (UAloss 0.03–0.09 kW/K)
sensor lag 3–6 s		[assumed]	—
process heat q_run 7–17 kW		[derived/assumed]: about 40 % of the 89 kW compound heating power of the project-12 extruder [R1] shared over its zones; mixer wall share assumed	envelope 0.5–1.5 × q_run
cooling water 8–30 °C		[assumed] cooling-tower circuit	an envelope axis
fouling Rf 0.000176 m²K/W, U_clean 5000 W/m²K		[std] TEMA order of magnitude [H3]; U [assumed]	clean vs fouled
mixer cycle 150 s, rotors loaded 65 %, ±0.6 q_run		[assumed]	the
Ms targets 1.4 / 1.6		[std] [C1]	—

8.3 Why "in some cases" (finding)

The model makes the cause of "in some cases" concrete. Around the loop's crossover the PID sees the cooling path as `b = dPc/dvalve = (dG/dvalve)·(Tw − Tcw)` together with an extra conductance `G(valve)` that speeds up the water mass. Both move:

- **Tw − Tcw** spans 13–82 K over setpoint 50–90 °C and cooling water 8–30 °C, a factor of about 6;
- **dG/dvalve** of the valve + exchanger falls with the opening on a linear (PWM) valve, because ε of the exchanger saturates as the cooling flow rises; on an equal-percentage valve the rising valve slope partly cancels that (`docs/img/exchanger_characteristic.png`).

The product moves by 13.7–25 × (linear valves) and 3.5–3.6 × (equal-percentage) relative to the test point (`results/study.md` §4). A PI tuned at the test point therefore meets at least one corner where it is either aggressive (winter, hot setpoint, low opening) or sluggish (summer, cool setpoint). The heating path has neither effect, which is why the same zone looks fine while it heats.



8.4 Plant set for design

The four zones are four different machine parts on purpose: two sensor positions (water, steel), two valve types, Cm 87–414 kJ/K, dead time 5.7–14.2 s, process heat 7–17 kW. The tool is tuned on each by its own simulated tests, so the result is a statement about the method across "similar TCUs", which is what the topic asks for.

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9. Reliability, failure modes and the FMEA ([W] ch. 34)

9.1 What counts as a failure

A failure of the zone control is any event that breaks SPS-101…105: an oscillation (limit cycle), a deviation beyond the limits, or an actuator held at a limit. Hardware failures of the TCU (heater, pump) are handled by the TCU's own protection and are out of scope; their effect on the controller is listed.

9.2 FMEA ([W] ch. 34; severity S, occurrence O, detection D on 1–10 scales, RPN = S·O·D)

ID	Failure mode	Effect	Cause	S	O	D	RPN	Mitigation in this design	RPN after
F1	PID in cooling with one gain set	limit cycle, valve and contactor wear	cooling path several times stronger than heating near crossover	6	7	5	210	crossover ratio (C) or linearisation (E)	60
F2	cooling gain drifts with season and setpoint	slow or oscillating zone "in some cases"	Pc ∝ (Tw − Tcw); valve and exchanger nonlinear in the opening	5	7	7	245	FB_CoolLinearise mode 2	50
F3	cooling-water sensor fails	wrong valve demand	wire break, AI fault	6	3	3	54	hold the last valid Tcw (`TcwValid`), alarm	36
F4	valve saturates before the PID's own limit	integrator windup, overshoot after a load drop	linearised demand above exchanger capacity	5	4	6	120	conditional integration (`CoolSaturated`)	40
F5	exchanger fouls	less cooling gain, then loss of capacity	scale, biofilm	4	6	6	144	the robustness margin covers it (Ms only falls); time in `Saturated` as a cleaning indicator	72

ID	Failure mode	Effect	Cause	S	O	D	RPN	Mitigation in this design	RPN after
F6	wrong nameplate data in FB_CoolLinearise	gain error	entry error; valve replaced by another type	5	3	6	90	`tcu-tune nameplate` print-out; the cooling step through the linearisation measures the result	45
F7	zone sensor fails	wrong PV	Pt100 break	8	3	2	48	existing TCU limit thermostat; PV plausibility (TBD-5)	—
F8	feedforward signal wrong	offset or extra disturbance	drive signal scaling	4	3	5	60	the PID corrects it; ±30 % error is still better than none (section 0)	48
F9	heater SSR stuck on	overheating	SSR failure	9	2	3	54	TCU over-temperature cut-out (hardware), out of scope	—

9.3 Availability

The software adds no new single point of failure except the Tcw sensor, and its failure (F3) degrades to "temperature scaling frozen", not to a stop. Automatic-mode availability therefore stays set by the existing TCU hardware.

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10. Analysis of alternatives ([W] ch. 32)

The AoA follows [W] ch. 32: alternatives, criteria and weights fixed first, utility curves stated, then a sensitivity check. Script: `scripts/aoa.py`, output `results/aoa.md`.

10.1 Alternatives

Key	Cooling path	New hardware	Engineering per TCU
A	one gain set (CoolGainRatio = 1)	none	none
B	static gain ratio from the cooling step	none	1 test
C	ratio at the crossover frequency (tool v0.1)	none	1 test
D	C + valve scaled by DTref / (Tw − Tcw)	cooling-water Pt100	sensor + test
E	C + valve + exchanger model inversion (tool v0.2)	cooling-water Pt100	sensor + nameplate data + test

10.2 Criteria, weights and results

Criterion (weight)	Measure	A	B	C	D	E
robustness (30)	worst Ms over the envelope, clean and fouled	298	149	1.70	3.35	1.41
corner performance (25)	mean load-step peak / SP-down overshoot at the corners (B, D: test point only)	2.49 / 1.80 K	1.77 / 2.58 K	4.14 / 0.55 K	2.70 / 0.75 K	3.26 / 0.54 K
consistency (15)	mean Ms spread (max − min) per zone	57.9	65.9	0.48	0.71	0.04
actuator wear (10)	heater/valve switchovers per run	129	80	26	26	39
cost (15)	sensor + engineering per TCU (assumed 250 EUR sensor, 60 EUR/h)	0	0	240	610	970 EUR
PLC load (5)	budget §7	low	low	low	low	0.64 ms/zone
figure of merit		**33.7**	**35.5**	**77.4**	**43.9**	**80.2**

The exact measures and utility scores are in `results/aoa.md`. A's low load-step peaks come from its aggressive cooling gain and are paid for with limit cycles and SP-down overshoots of up to 6.4 K. Corner performance scores C and E almost equal on average, because the head zone (sensor in the steel) dominates both means; the separation is at the summer corner of the extruder zones (2.9 K for C vs 1.1–1.7 K for E).

10.3 Sensitivity

- Every weight ±10 points, one at a time: E wins 10 of 12 cases; C wins when consistency loses 10 points or PLC load gains 10 (by 0.1).

- Cost doubled: C wins (76.2).
- Cost weight at which E stops winning: 27 of 100.

10.4 Recommendation

The ranking is close and the reason is clear: E buys uniform robustness with one sensor per TCU and the exchanger data; C already removes the instabilities for free. Recommendation, in order:

1. retune every zone with the tool's crossover ratio (C) — no hardware, removes A's limit cycles;
2. add the cooling-water sensor and E on TCUs whose cooling water swings with the season or whose valves are equal-percentage, where C becomes slow;
3. add power feedforward on mixer and head zones (section 0, item 6);
4. design a cascade on the water temperature for zones with the sensor in the steel (section 13).

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11. Verification and validation ([W] ch. 13, 28)

11.1 Requirements verification traceability matrix ([W] Table 22.3, p.458)

The full matrix with code and test locations is in `TRACEABILITY.md`.

Req.	Method	Result (simulation)	Status
SPS-101 ±1 K	A + T	not evaluated as a time share: the study runs scenarios, not production weeks	open (VAL-1)
SPS-102 load step ≤ 2 K	A	E: 0.9–1.7 K on zones A, B; 1.8–2.2 K on the mixer zone D; head zone C 6.5 K (2.6 K with feedforward)	met on A, B; D misses at the corners by 0.2 K; C not met (section 13)
SPS-103 SP-down ≤ 1 K	A	C: 0.0–0.9 K; E: 0.2–0.7 K on A, B, D and 1.2 K on the head zone C; A: up to 6.4 K	C met; E met except the head zone; A, B not
SPS-104 no limit cycle	A	A: 707 switchovers with the valve slamming 0–100 % (zone A test point); C, E: 2–122 sign changes per run from noise near zero output, no valve cycling	C, E met; A, B not
SPS-105 batch ripple ≤ ±0.5 K	A	±0.4 K without, ±0.2 K with feedforward	met
SPS-201 Ms ≤ 1.6 over the envelope	A	E 1.36–1.41 (fouled 1.19–1.33); C up to 1.61–1.70	E met; C misses by 0.01–0.10 at a few corner points of every zone
SPS-202/203 tool	D, T	pytest `test_cli.py`, `test_rules_reject_missing_dead_time`	met
SPS-204/205 bumpless, conditional integration	T	pytest, SCL bit-exact	met
SPS-206 Tcw fault hold	T	pytest on the SCL	met
SPS-301/302 signals	I	section 6.2	met
SPS-361 TIA V18 0/0	T	0 errors, 0 warnings incl. OB30	met
SPS-362 ≤ 20 % of OB30	A	15 % (upper bound, 8 zones, 3× margin)	A met; T open (VT-7)
SPS-363 SCL = reference	T	bit-exact / ≤ 0.01 % / ≤ 0.05 K	met

11.2 What has been verified in this repository

- **Unit and equivalence tests** (`pytest`, run in CI on every push):
  - physics: water table, ε-NTU limits, valve end points, Gnielinski range, derived presets physical;

    - plant: steady state, Euler accuracy, linearisation; identification and the rules; the tool end to end;
    - the PID, including conditional integration;
    - the SCL source executed in float32 against the Python references: `FB_ZonePID` bit-exact with manual, reset, feedforward and the saturation flag; `FB_CoolLinearise` within 0.01 % valve in all three modes on three zones; the whole SCL loop (`FB_ZoneTest`) within 0.05 K of the Python loop, with and without linearisation and feedforward.

- **TIA Portal V18 compile** through Openness, CPU 1513-1 PN FW 2.9: all FBs, the DBs and OB30 at 100 ms, 0 errors, 0 warnings (`results/tia_compile*.txt`, `docs/img/tia_FB_CoolLinearise.png`).
- **Analysis** (`scripts/run_study.py`): envelope Ms, corners in the time domain, batch load, budget.
- **Not verified here:** a PLCSIM run (S7-PLCSIM V18 is not installed; its installer is on the PC but stops on a pending Windows restart), execution time on a real CPU, any real TCU.

11.3 Line-trial plan (validation) ([W] P13.13, p.284/314; ch. 28)

The acceptance criteria are set before the trial ([W] P13.13).

Step	What	Pass criterion
VT-1	Record the current settings and one week of trends for 4 zones, one of each type	data complete
VT-2	Enter the nameplate data (`tcu-tune nameplate` format), install the Tcw sensor	reading within ±0.5 K of a reference thermometer
VT-3	Heater step, cooling step (through FB_CoolLinearise) and relay test per zone	tool report produced; every FOPDT refusal explained
VT-4	Load the recommended settings into the existing FB on one zone; compare a load change and a setpoint step with VT-1	SPS-102, 103 met, or better than the current settings
VT-5	Switch on FB_CoolLinearise mode 2; repeat VT-4 in a cold-water week and a warm-water week	switchovers within SPS-104; no slow response in the warm week
VT-6	Mixer zone: feedforward on and off on alternate days	ripple at the batch period reduced
VT-7	CPU cycle statistics with all zones	SPS-362
VAL-1	Process engineering compares compound-temperature records over one month	SPS-101 met, accepted by the independent tester ([W] P13.15)

12. TBD register and questions for the plant

ID	Item	Why it matters	Who
TBD-1	The real temperature tolerance per zone (±1 K assumed)	sets SPS-101/102	process engineering
TBD-2	Cooling-water circuit: tower or chiller, temperature range over the year	the envelope axis behind F2	utilities
TBD-3	TCU type, exchanger data sheet (UA or kW at given flows), valve type (PWM solenoid or proportional, Kvs, characteristic)	FB_CoolLinearise inputs	maintenance, supplier
TBD-4	The current PID settings and how the cooling output is scaled today (one gain set?)	baseline for VT-4	automation
TBD-5	Is the zone sensor in the water or in the steel; is the TCU water temperature available in the PLC	lead-lag vs FOPDT; the Tw input	automation
TBD-6	Which process-power signals exist (extruder drive, mixer rotor power)	feedforward	automation
TBD-7	Which zones show the "non-optimal behaviour", and when (season, product)	confirms or rejects the envelope explanation	operators
TBD-8	Trends of heater and valve outputs during an episode	tells a limit cycle (F1) from a slow response (C in summer)	automation

13. Limits of this work

- **Simulation only.** The hardware values are sizing assumptions for this machine class (section 8.2). The relations are textbook, but the numbers will move with the real TCUs; the method is what carries over.
- **Two lumped masses per zone.** A barrel zone has an axial temperature profile and couples to its neighbours; here the neighbours are a fixed conductance.
- **The head zone C (sensor in the steel)** misses SPS-102 with every cooling alternative: a 14 kW load step goes straight into the steel where the sensor is, and the water can only act through the steel-to-water lag. Feedforward from the process power cuts the deviation (section 0), but the peak stays above 2 K. A cascade on the TCU water temperature is the next design step and is not built here (decision D6). A real head load also rises with screw speed rather than as a step.
- **The linearisation needs the exchanger model.** The fouled case shows it degrades gracefully (Ms only falls), but a wrong valve type in the nameplate data would bias the gain (F6).
- **PLCSIM and on-target timing** are not done; the budget is a static upper-bound count.
- **No real line data.** All findings are hypotheses for the line trial (11.3).

14. Decision log

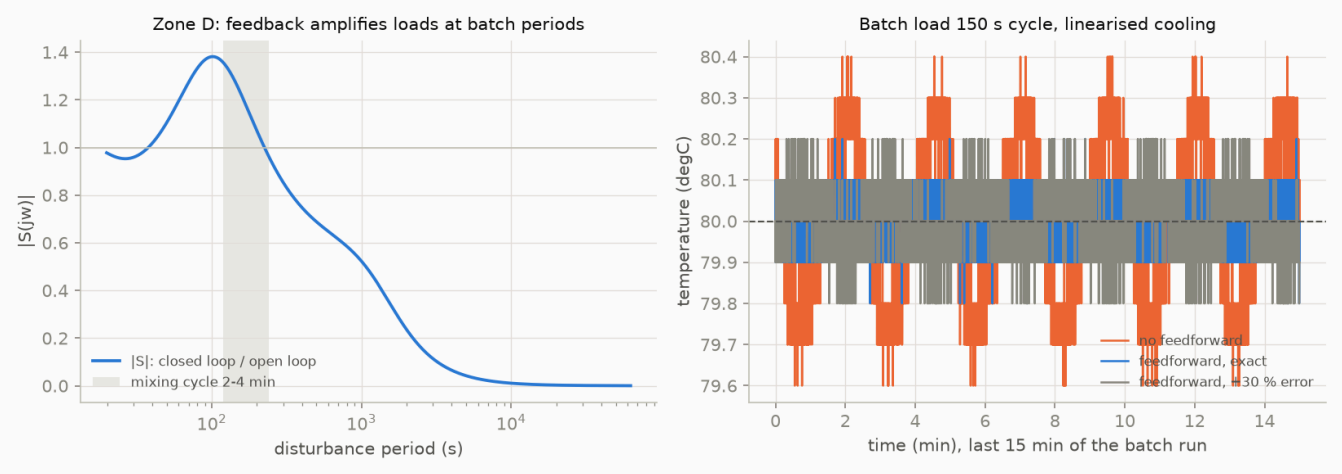
ID	Decision	Alternatives considered	Reason
D1	Derive the zone parameters from hardware (`physics.py`)	keep the v0.1 invented presets	the plant can replace each assumed value with a nameplate value; the method then carries over
D2	Model the cooling as valve + plate exchanger (ε-NTU) in a closed primary circuit	bilinear direct-cooling law	the topic describes a closed primary circuit; ε-NTU makes the valve gain depend on the opening, the second cause of "in some cases"
D3	Linearise outside the PID, in a new FB	gain-schedule the PID parameters	the plant's PID block stays standard; valve and exchanger knowledge belongs to the TCU type and is entered once
D4	Bisection with 16 model evaluations for the inversion	lookup table; Newton	G(v) is monotonic, so bisection cannot fail and has a fixed cost; a table needs a staircase test per TCU
D5	Conditional integration from the linearisation's saturation flag	ignore it	in the summer corner the valve saturated before the PID's own limit and the integrator wound up
D6	No cascade for the steel-sensor zone in this revision	cascade on Tw	outside the cooling-path question; recorded as the next step (section 13)
D7	`100.0 * 0.5 * (…)` rewritten as `50.0 * (…)`	—	TIA V18 folds the constant product into an LReal and warns that accuracy may be lost (warning in line 78); found by compiling six variants of the block
D8	Hold the last Tcw on a sensor fault	fall back to mode 0	mode 0 needs a different CoolGainRatio; Tcw moves over hours, so holding it is safe

15. References

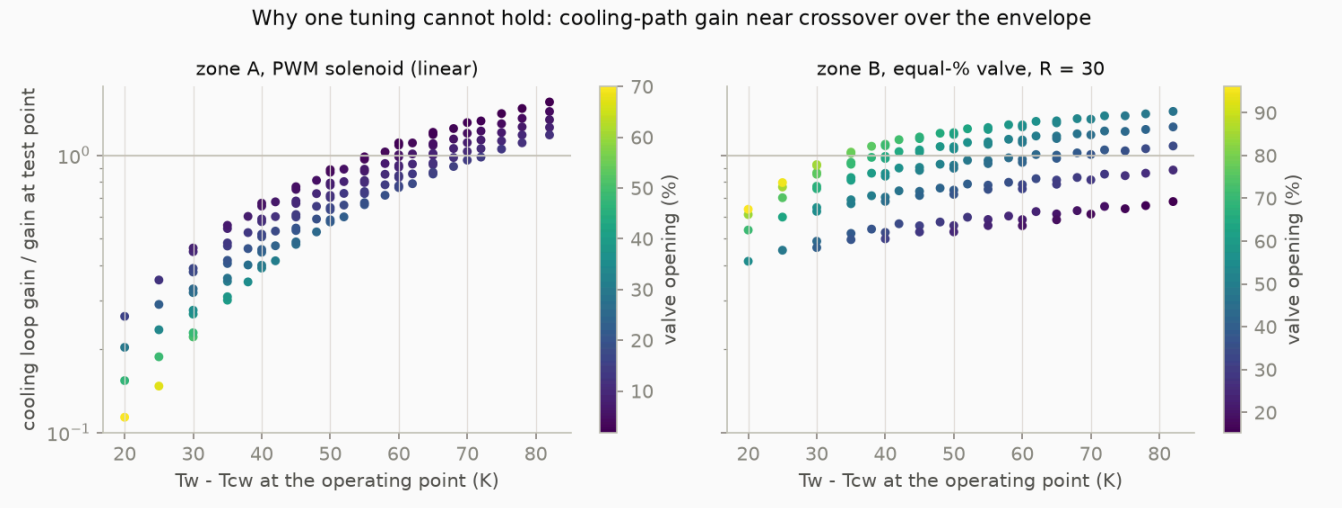
See `docs/sources/REFERENCES.md` for the full list with source status. Short keys used here: [W] Wasson; [H1] Bergman/Incropera heat transfer; [H2] Shah & Sekulić; [H3] TEMA fouling table; [V1] ISA-75.11.01; [R1] project 12 extruder model; [R2] White, rubber processing; [C1] Åström & Hägglund; [C2] Skogestad SIMC; [C3] Skogestad & Postlethwaite; [C4] Shinskey; [C6] Ljung; [P1] CPU 1513-1 PN data sheet; [P2] TIA Portal V18.

Appendix: figures

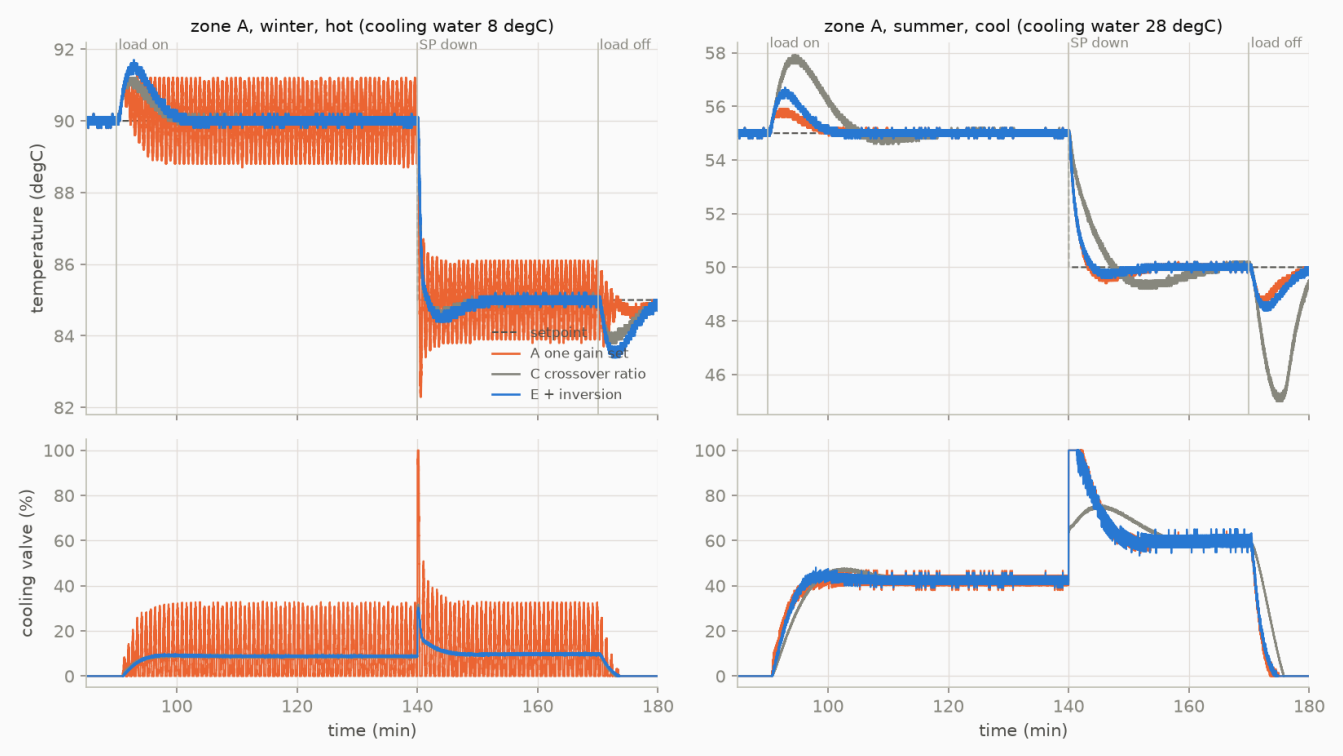
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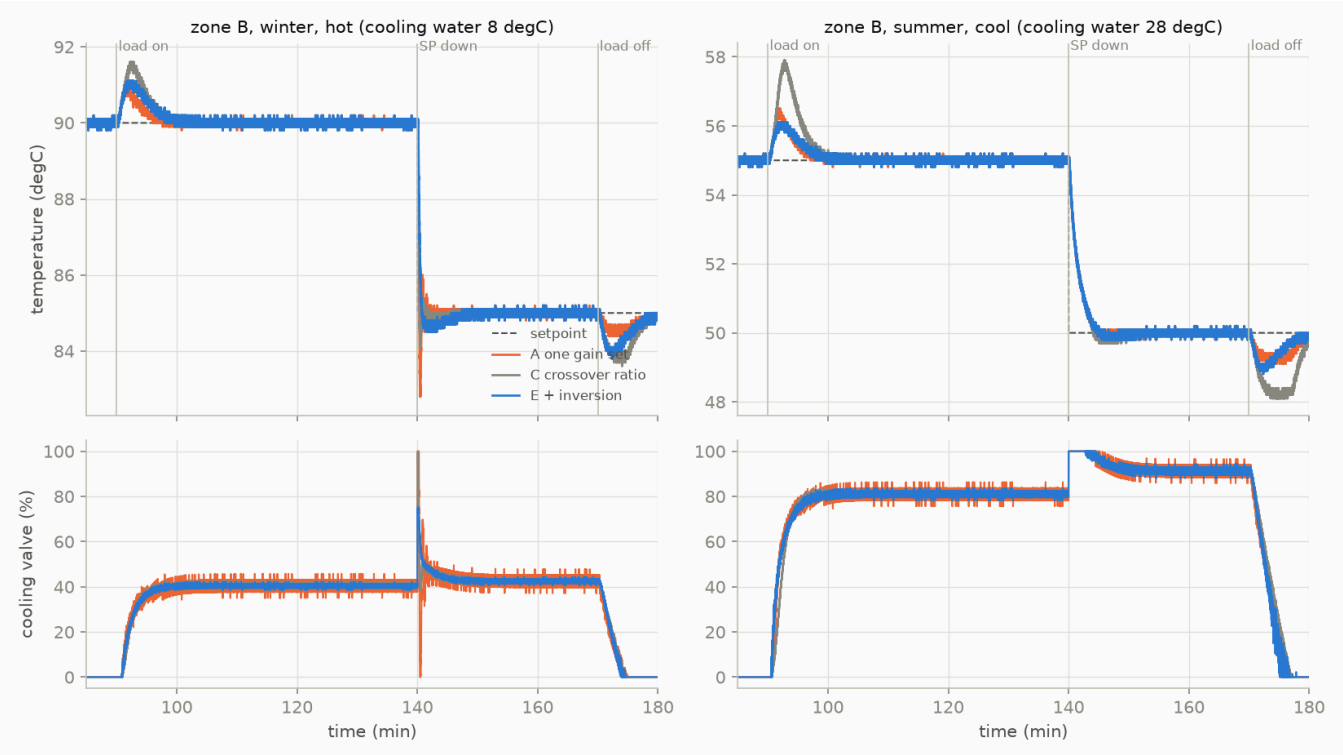
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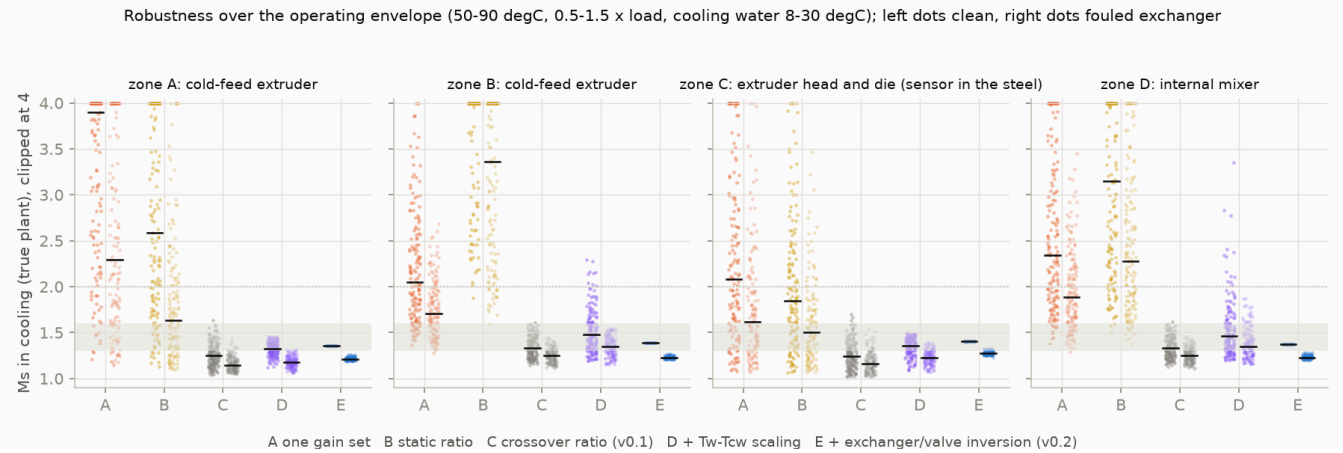
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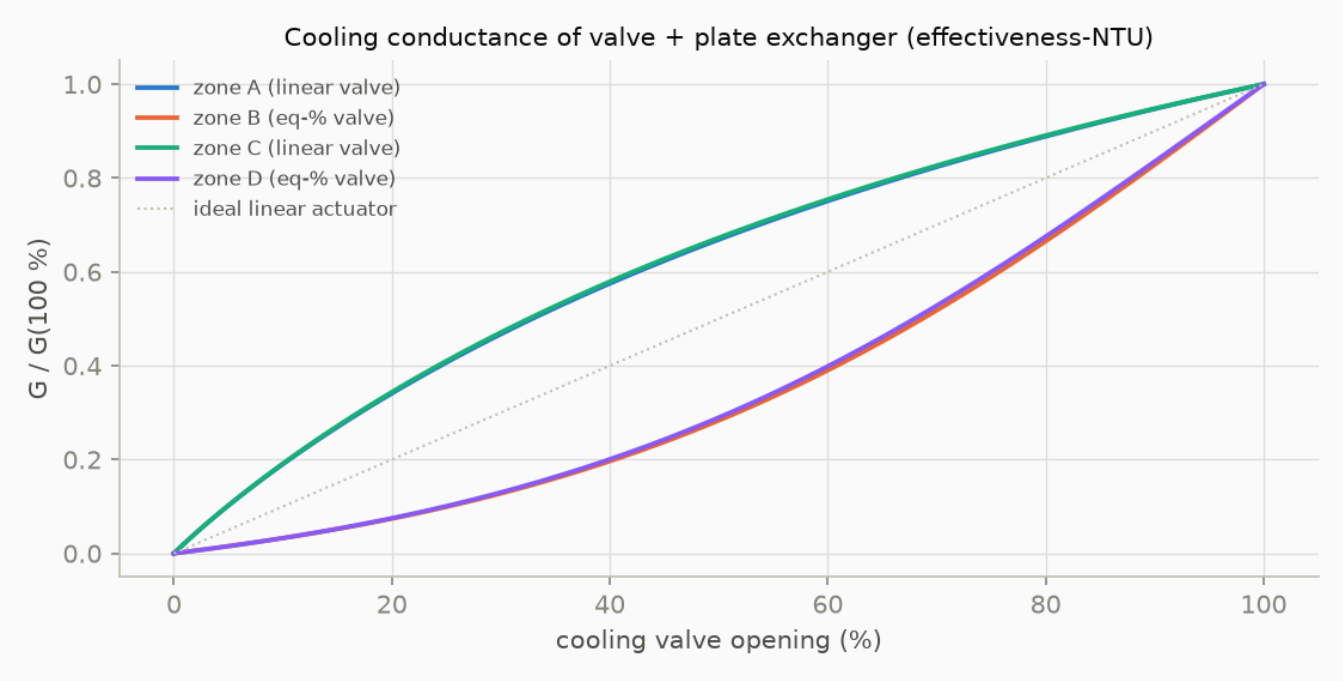
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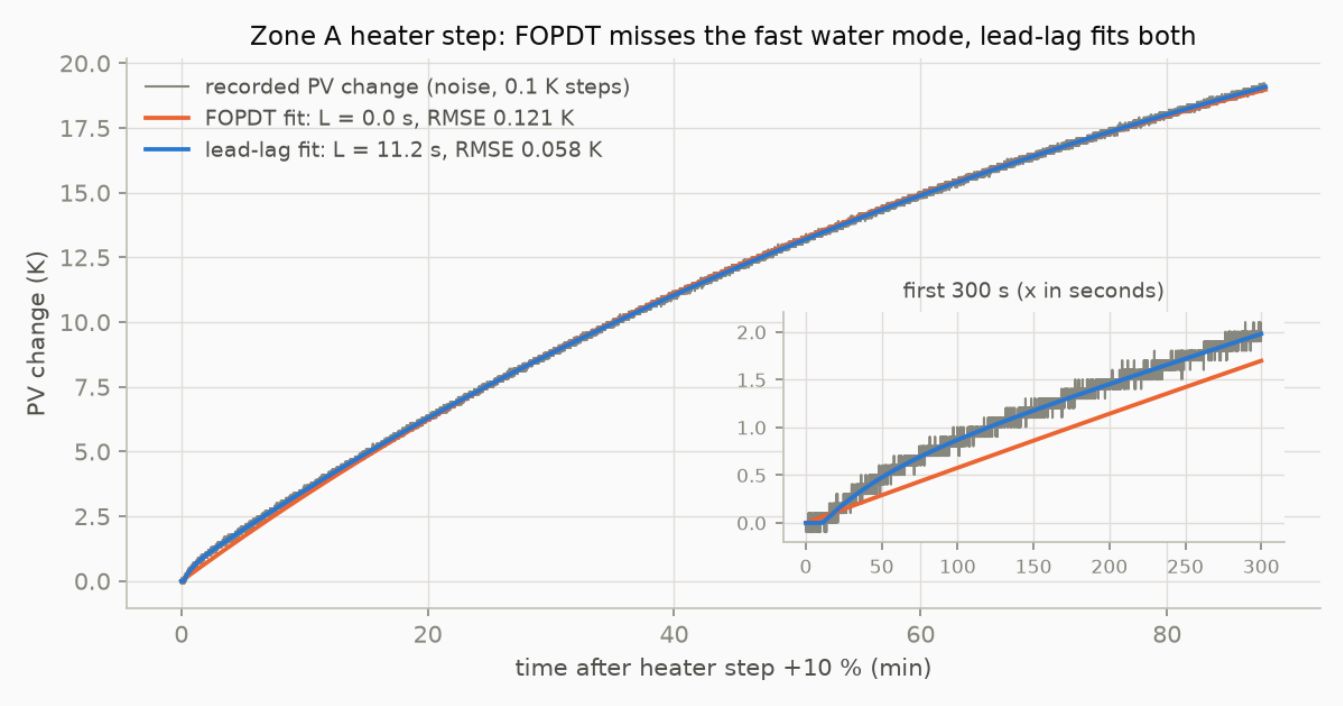
envelope_ms.png



exchanger_characteristic.png



identification_zoneA.png



tia_FB_CoolLinearise.png

